Lifecycle Transition Control Module (LTCM)

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OOF® Origin Open Foundation®

Independent Methodological Authority

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Subcategory: Software Lifecycle Control Governance

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Governed Space: Software Lifecycle Transition Governance

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Canonical Definition

Lifecycle Transition Control Module (LTCM) is the Core Module within Software Lifecycle Control Standard (SLCS) that governs how identifiable software objects legitimately transition between lifecycle states through defined conditions, applicable requirements, competent authority, bounded execution, verifiable outcomes, and preserved governance continuity.

LTCM establishes lifecycle transition control as an evidence-supported

governance relationship between an established source state, an intended destination state, the requirements governing movement between them, the actors authorized to initiate or execute the transition, and the actual state achieved.

The module distinguishes a proposed transition from an authorized

transition, technical transition capability from legitimate authority, execution completion from verified state achievement, and successful state change from compliance with the conditions under which the change occurred.

It requires material lifecycle transitions to remain associated with

identifiable software objects, applicable versions, configurations, environments, source and destination states, transition purposes, mandatory preconditions, authority boundaries, execution evidence, verification results, and unresolved obligations.

LTCM recognizes that software lifecycles are not necessarily linear. Transitions may occur between parallel lifecycle dimensions, involve partial scopes, return software to earlier governance activities, or establish temporary, restricted, conditional, or exceptional states.

The module therefore does not impose a universal sequence of lifecycle stages.

Instead, it governs whether a particular transition is legitimate within the actual lifecycle context of the software being governed.

LTCM also recognizes that a technically successful transition may remain unauthorized, insufficiently verified, or inconsistent with applicable governance requirements.

Its principal governance output is the Software Lifecycle Transition Control Profile.

Operational Role

LTCM establishes the governance mechanism for legitimate software lifecycle

transitions.

It governs:

• Lifecycle transition case identity

• Source lifecycle state

• Intended destination lifecycle state

• Transition origin

• Transition purpose

• Transition scope

• Transition classification

• Applicable governance requirements

• Transition preconditions

• Transition feasibility

• Transition readiness

• Transition authority

• Transition responsibility

• Execution permission

• Execution boundaries

• Transition sequencing

• Cross-standard dependencies

• Transition evidence

• Transition execution traceability

• Transition outcome verification

• Partial transitions

• Failed transitions

• Unauthorized transitions

• Conditional transitions

• Automated transitions

• Autonomous transition actions

• Transition interruption

• Transition recovery

• Transition rollback

• Transition deviations

• Transition exceptions

• Transition uncertainty

• Transition state reconciliation

• Transition closure

• Transition reassessment

LTCM governs the legitimacy and verifiability of movement between lifecycle

states.

It does not replace the specialized technical or methodological processes that establish the requirements of individual lifecycle domains.

Operational Question

How can an identifiable software object move between lifecycle states only when the applicable requirements, source conditions, authority boundaries, execution permissions, and verification obligations have been satisfied, even when transitions are automated, nonlinear, partial, or distributed across multiple systems and actors?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-Cutting Governance Layer

Software Lifecycle Control Standard (SLCS)

SLCS Core Module Flow

Lifecycle State Consistency Control Module (LSCCM) → Lifecycle Transition

Control Module (LTCM) → Lifecycle Exception Management Module (LEMM) → Lifecycle Control Transfer & Continuity Module (LCTCM) → Lifecycle Control Assessment Module (LCAM)

LTCM is the second Core Module within SLCS.

It receives evidence-supported lifecycle state information from LSCCM and

establishes the governance conditions required for legitimate transitions.

It interacts with LEMM where exceptional transition conditions arise.

It interacts with LCTCM where transitions affect responsibility, authority, or control continuity.

It provides transition control evidence to LCAM for the assessment of

governance effectiveness.

• Governed Space

Software Lifecycle Transition Governance

This governed space concerns the conditions under which software lifecycle

state changes may legitimately occur.

It begins when a material lifecycle transition is proposed, required, detected, or initiated.

It includes transition identification, source-state assessment,

destination-state definition, requirements, preconditions, authority, execution boundaries, verification, deviations, and resulting state classification.

It remains applicable where transitions are incomplete, interrupted, unauthorized, conditionally permitted, partially executed, or subject to reassessment.

It concludes for an individual transition when the actual outcome has been established and the applicable transition obligations have received an appropriate governance disposition.

Governed Object

The primary Governed Object is:

A Software Lifecycle Transition Case representing the evidence-supported

relationship between an identifiable software object, its established source lifecycle state, intended destination state, transition purpose, applicable requirements, mandatory preconditions, competent authority, execution conditions, actual transition activity, verification evidence, and resulting lifecycle state.

The governed object must preserve, where applicable:

Transition Case ID

Software Object ID

Software Version

Configuration Reference

Environment Reference

Source Lifecycle State

Intended Destination State

Transition Origin

Transition Purpose

Transition Scope

Applicable Requirements

Mandatory Preconditions

Transition Authority

Execution Responsibility

Execution Permission

Execution Evidence

Actual Resulting State

Verification Evidence

Transition Classification

Residual Obligations

Record Version

Scope

LTCM governs:

Transition Identification

Transition Purpose

Transition Scope

Source-State Verification

Destination-State Definition

Transition Requirement Identification

Transition Preconditions

Transition Readiness

Transition Authority

Transition Responsibility

Execution Permission

Transition Execution Boundaries

Transition Sequencing

Transition Dependencies

Transition Evidence

Transition Verification

Transition Failure

Partial Transition

Conditional Transition

Unauthorized Transition

Automated Transition

Autonomous Transition

Transition Recovery

Transition Rollback

Transition Deviation

Transition Exception Interface

Transition Closure

Transition Reassessment

LTCM does not prescribe one universal technical transition method.

Its governance requirements adapt to the software object and the transition

being performed.

Foundational Principle

A software lifecycle transition is legitimate only when the movement from an established source state to an intended destination state occurs under applicable requirements, competent authority, bounded execution conditions, and sufficient evidence of the actual outcome.

A technically possible transition is not necessarily permitted.

An authorized transition is not necessarily completed.

A completed transition is not necessarily verified.

And a verified resulting state does not retroactively legitimize an unauthorized execution.

Technical Transition Capability ≠ Legitimate Transition Authority.

Transition Executed ≠ Transition Verified.

Lifecycle Transition Case Identity

Every material lifecycle transition must have a stable case identity.

Relevant elements include:

Transition Case ID

Software Object Reference

Source State Reference

Destination State Reference

Transition Origin

Responsible Function

Transition Status

Evidence Reference

Record Version

The identity must remain stable across assessment, authorization, execution, verification, and closure.

Transition Origin

A lifecycle transition may originate from:

Requirements Completion

Development Progression

Quality Assessment

Security Assessment

Release Decision

Deployment Requirement

Operational Change

Dependency Change

Configuration Change

Incident Response

Support Transition

Retirement Requirement

Automated System Event

Autonomous Agent Action

The origin of a transition must not be confused with its authorization.

Transition Purpose

LTCM must establish why the transition is required.

Potential purposes include:

Lifecycle Progression

Software Activation

Software Restriction

Software Modification

Release Preparation

Deployment

Operational Recovery

Security Remediation

Migration

Retirement

Lifecycle Re-entry

Governance Correction

Transition purpose must remain connected to the intended destination state.

Source Lifecycle State

LTCM must establish the source state from which the transition is proposed or has occurred.

Relevant elements include:

Software Identity

Lifecycle Dimension

Current State

Applicable Scope

Version

Configuration

Environment

Verification Evidence

Known Limitations

The source state must be sufficiently established for the intended governance decision.

Source-State Validity

LTCM must evaluate whether the source state remains valid at the time of transition.

Relevant considerations include:

State Evidence

Temporal Applicability

Configuration Changes

Deployment Changes

Operational Changes

Unresolved Conflicts

Required Reassessment

A previously verified source state may become outdated before transition execution.

Intended Destination State

LTCM must define the intended destination state.

Relevant elements include:

Destination State Identity

Lifecycle Dimension

Expected Operational Condition

Applicable Scope

Required Verification Criteria

Mandatory Governance Conditions

Continuing Obligations

The intended destination must be sufficiently defined to distinguish successful transition from incomplete execution.

Destination-State Requirements

The destination state may require:

Required Artifacts

Required Configuration

Quality Conditions

Security Conditions

Release Authorization

Deployment Readiness

Operational Verification

Dependency Suitability

Retirement Verification

LTCM identifies the applicable requirements without replacing the

specialized standards that establish them.

Transition Scope

Every material transition must have a defined scope.

Potential dimensions include:

Software Product

Software Version

Build

Configuration

Component

Deployment Instance

Environment

Operational Capability

User Group

Geographic or Organizational Scope

Transition outcomes must not be generalized beyond the verified scope.

Transition Boundary

LTCM must establish what is included and excluded from the transition.

Relevant elements include:

Affected Software Objects

Affected Lifecycle Dimensions

Affected Environments

Included Dependencies

Excluded Dependencies

Affected Actors

External Systems

Continuing Conditions

A transition boundary must remain consistent with the actual operational relationships affected.

Transition Classification

Transitions may be classified as:

Planned

Required

Conditional

Routine

Emergency

Automated

Autonomous

Partial

Re-entry

Recovery

Rollback

Retirement

Exceptional

Classification must not replace assessment of actual requirements and authority.

Lifecycle Transition Preconditions

LTCM must identify mandatory conditions that must be satisfied before transition execution.

Potential preconditions include:

Source State Established

Destination Requirements Defined

Required Assessments Completed

Applicable Evidence Available

Dependencies Evaluated

Required Resources Available

Operational Conditions Suitable

Competent Authority Established

Verification Method Defined

Mandatory preconditions must not be bypassed through administrative status changes.

Precondition Verification

Preconditions must be evaluated through applicable evidence.

Relevant elements include:

Precondition Identity

Verification Method

Evidence

Responsible Function

Verification Time

Result

Known Limitations

Unresolved Conditions

A precondition marked complete without appropriate evidence must not automatically be treated as satisfied.

Transition Feasibility

LTCM must assess whether the intended transition can be performed under the actual software and operational conditions.

Relevant dimensions include:

Technical Feasibility

Operational Feasibility

Dependency Conditions

Resource Availability

Required Capability

Security Conditions

Recovery Requirements

External Constraints

Feasibility does not establish authorization.

Transition Readiness

Transition readiness concerns whether the necessary conditions are established for execution.

Relevant elements include:

Verified Preconditions

Required Resources

Responsible Actors

Execution Method

Operational Window

Dependency Readiness

Verification Capability

Recovery Preparedness

Readiness must remain distinguishable from the authority to proceed.

Transition Authority

LTCM must identify competent authority for material lifecycle transitions.

Relevant elements include:

Authority Holder

Authority Origin

Authority Scope

Permitted Transition

Delegation

Validity Conditions

Execution Restrictions

Revocation Conditions

External Approval Requirements

Authority must remain applicable to the specific transition and affected software scope.

Authority Scope

Transition authority may be limited by:

Software Object

Version

Environment

Lifecycle Dimension

Transition Type

Operational Risk

Time Window

Execution Mechanism

Required Oversight

Authority for one transition must not automatically authorize another.

Transition Decision

A transition decision must preserve:

Transition Identity

Decision Origin

Applicable Requirements

Precondition Status

Evidence Basis

Competent Authority

Decision

Conditions

Decision Time

Validity

A transition decision may authorize, reject, defer, or condition execution.

Transition Responsibility

LTCM must identify responsibility for:

Transition Assessment

Precondition Verification

Transition Decision

Execution

Operational Monitoring

Outcome Verification

Deviation Management

Recovery

Closure

Responsibility must remain aligned with actual capability and applicable authority.

Execution Permission

Execution permission establishes whether a particular actor or system may perform the transition action.

Relevant elements include:

Actor Identity

Permitted Action

Software Scope

Environment

Tool or Mechanism

Operational Constraints

Authority Reference

Validity Conditions

Technical access must not substitute for execution permission.

Execution Boundary

LTCM must establish the limits within which transition execution may occur.

Potential boundaries include:

Authorized Software Objects

Permitted Environments

Permitted Configuration Changes

Permitted Deployment Targets

Allowed Tools

Allowed Time Window

Maximum Operational Impact

Mandatory Intervention Conditions

Execution outside the applicable boundary must remain identifiable.

Transition Sequencing

Some transitions require ordered activities.

Relevant elements include:

Prerequisite Activities

Dependency Sequence

Execution Order

Verification Points

Intervention Points

Recovery Conditions

Completion Criteria

Sequencing must reflect the actual transition rather than impose unnecessary universal steps.

Cross-Standard Transition Dependencies

Transitions may depend on outcomes governed by multiple Parent Standards.

Relevant relationships include:

Requirements Completion

Creation Evidence

Dependency Suitability

Configuration Baseline

Quality Assessment

Security Assessment

Release Authorization

Deployment Readiness

Change Coordination

Retirement Readiness

LTCM must preserve these dependencies without independently replacing their specialized decisions.

Transition Evidence

Material transitions must preserve appropriate evidence.

Potential evidence includes:

Source-State Verification

Precondition Records

Authorization Records

Execution Logs

Configuration Changes

Deployment Records

Operational Observations

Verification Results

Recovery Records

Closure Decisions

Evidence must remain associated with the transition scope and applicable time.

Transition Execution Traceability

LTCM must preserve traceability between authorization and execution.

Relevant elements include:

Authorized Transition

Executing Actor

Execution Mechanism

Execution Time

Affected Software

Actual Actions

Observed Changes

Evidence

Deviations

Traceability must permit reconstruction of material transition activities.

Transition Execution State

A transition may occupy states including:

Proposed

Assessment Pending

Preconditions Pending

Ready

Authorized

Execution Active

Interrupted

Partially Executed

Executed

Verification Pending

Verified

Failed

Rejected

Closed

Unknown

A transition execution state must not be confused with the software's resulting lifecycle state.

Transition Outcome Verification

LTCM must verify the actual result of a transition.

Relevant elements include:

Intended Destination State

Actual Resulting State

Applicable Verification Criteria

Verification Method

Evidence

Responsible Function

Known Deviations

Resulting Classification

Verification must establish what actually changed.

Verification Evidence Applicability

Transition verification evidence must remain applicable to:

Software Identity

Version

Configuration

Environment

Transition Scope

Execution Time

Verification Time

Intended Outcome

Evidence from another scope must not automatically establish transition success.

Transition Outcome Classification

LTCM recognizes the following outcome classifications:

Transition Verified — Intended State Achieved

Transition Verified — Partial State Achieved

Transition Executed — Verification Pending

Transition Failed

Transition Interrupted

Transition Unauthorized

Transition Conditionally Valid

Transition Deviation Identified

Transition Reassessment Required

Transition Outcome Unknown

Classification must preserve both execution legitimacy and resulting state evidence.

Partial Transition

A partial transition occurs when only part of the intended scope reaches the destination state.

Relevant elements include:

Completed Scope

Incomplete Scope

Actual Resulting States

Operational Consequences

Continuing Requirements

Required Actions

Verification Evidence

Partial completion must not be represented as complete transition.

Failed Transition

A failed transition occurs when the intended outcome is not achieved or mandatory conditions are not satisfied.

Potential conditions include:

Execution Failure

Destination State Not Achieved

Mandatory Precondition Violation

Operational Failure

Verification Failure

Unrecoverable Deviation

Failure must remain distinguishable from a transition that was never authorized.

Unauthorized Transition

An unauthorized transition occurs when a material lifecycle state change is executed without applicable legitimate authority.

Relevant elements include:

Affected Software

Executing Actor

Executed Action

Authority Gap

Actual Resulting State

Operational Consequences

Required Response

Evidence

A technically successful unauthorized transition must not be reclassified as authorized solely because its intended result was achieved.

Conditional Transition

A conditional transition may be permitted under explicitly established conditions.

Relevant elements include:

Transition Identity

Conditional Authority

Applicable Restrictions

Required Controls

Residual Risk

Verification Requirements

Validity Conditions

Reassessment Triggers

Conditional transition must not be used to conceal unmet mandatory requirements.

Transition Interruption

A transition may be interrupted before its intended destination state is established.

Potential causes include:

Technical Failure

Operational Intervention

Security Event

Dependency Failure

Authority Withdrawal

Resource Loss

Unexpected Operational Condition

The interrupted state must remain identifiable and governed.

Transition Recovery

Transition recovery concerns restoring an appropriate governed condition

after interruption or failure.

Relevant elements include:

Failure State

Recovery Objective

Applicable Authority

Recovery Preconditions

Recovery Method

Operational Consequences

Verification Evidence

Resulting State

Detailed recovery execution remains governed by the applicable specialized

lifecycle mechanisms.

Transition Rollback

Rollback may return software toward an earlier state.

LTCM must preserve:

Rollback Origin

Rollback Authority

Target State

Applicable Preconditions

Affected Software

Execution Evidence

Verification Evidence

Residual Conditions

Rollback must be governed as a transition rather than assumed to restore every earlier condition automatically.

Transition Reversibility

LTCM must recognize whether a transition can be reversed.

Relevant elements include:

Reversible Conditions

Irreversible Changes

Data Transformations

External Effects

Dependency Changes

Authority Changes

Recovery Capability

Operational Constraints

Reversibility must be assessed rather than inferred from the availability of a rollback mechanism.

Lifecycle Re-entry Transition

A lifecycle re-entry transition returns software to an earlier or

previously completed governance domain.

Potential triggers include:

Requirements Revision

Quality Failure

Security Finding

Dependency Change

Configuration Change

Operational Incident

Retirement Reassessment

LTCM governs transition legitimacy and resulting lifecycle state.

Detailed re-entry control remains within SCMS and applicable specialized standards.

Automated Lifecycle Transition

Automated systems may execute lifecycle transitions under established authority.

Relevant elements include:

Automation Identity

Authorized Transition Type

Execution Conditions

Precondition Verification

Tool Permissions

Operational Boundaries

Execution Evidence

Outcome Verification

Automation must not create its own authority through technical capability.

Autonomous Lifecycle Transition

Autonomous agents may propose, select, or execute lifecycle transitions.

LTCM must preserve:

Agent Identity

Assigned Objective

Applicable Authority

Permitted Transition

Permitted Tools

Execution Path

Operational Constraints

Human Intervention Conditions

Evidence

Accountability

The authority to achieve an objective does not automatically authorize every constituent action.

Task Authority ≠ Constituent Action Authority.

Tool-Path Transition Authority

Where lifecycle transitions involve multiple tools or downstream systems, LTCM must preserve the authority applicable to material execution paths.

Relevant elements include:

Transition Objective

Authorized Actor

Tool Selection

Tool Permission

Target System

Data Access

Execution Action

Operational Effect

Verification Evidence

Authorization of the overall transition must not be treated as unrestricted authorization of every tool path used to execute it.

Human Oversight and Intervention

Where required, LTCM must establish effective human oversight.

Relevant elements include:

Oversight Function

Intervention Authority

Intervention Trigger

Execution Visibility

Pause Capability

Override Capability

Escalation

Intervention Evidence

Human oversight must represent a practical control capability rather than a nominal assignment.

Transition Deviation

A transition deviation occurs when actual execution or resulting conditions differ materially from applicable requirements.

Potential deviations include:

Unauthorized Action

Unexpected Configuration

Unverified Destination State

Execution Outside Scope

Missing Mandatory Evidence

Dependency Failure

Operational Disruption

Unapproved Tool Path

Deviations must remain identifiable and subject to appropriate governance.

Transition Exception Interface

Where a transition requires departure from ordinary lifecycle requirements, LTCM must preserve the relationship to LEMM.

Relevant elements include:

Affected Requirement

Exception Identity

Competent Authority

Permitted Deviation

Alternative Controls

Validity Conditions

Residual Risk

Verification Requirements

LTCM does not independently legitimize exceptions.

Transition Uncertainty

LTCM must preserve uncertainty where transition conditions or outcomes cannot be established.

Potential conditions include:

Unverified Source State

Unknown Execution Result

Incomplete Deployment Evidence

Conflicting Operational Observations

Unclear Authority

Unknown Dependency Impact

Missing Verification

Material uncertainty must not be converted into a definitive successful transition status.

Transition Monitoring

Some transitions require monitoring during or after execution.

Potential monitoring concerns include:

Execution Progress

Operational Availability

Deployment State

Dependency Behavior

Security Conditions

Performance Conditions

Unexpected Actions

Recovery Triggers

Monitoring requirements must remain proportionate to the governed transition.

Transition Reassessment

LTCM must support reassessment when:

Source State Changes

Destination Requirements Change

Authority Changes

Preconditions Expire

New Dependency Evidence Appears

Security Conditions Change

Execution Is Interrupted

Verification Fails

Material Deviation Is Identified

Reassessment must preserve earlier decisions and their evidence.

Transition Closure

A transition case may be closed when its actual outcome and continuing obligations have received an appropriate governance disposition.

Closure must preserve:

Transition Identity

Authorized Scope

Actual Execution

Resulting State

Verification Evidence

Deviations

Residual Obligations

Closure Decision

Transition closure does not independently establish the completion of every broader software lifecycle activity.

Transition Governance Classification

LTCM recognizes the following governance classifications:

Transition Proposed

Assessment Pending

Preconditions Pending

Readiness Verified

Authorization Pending

Authorized

Conditionally Authorized

Rejected

Deferred

Execution Active

Interrupted

Partially Executed

Executed — Verification Pending

Verified Complete

Verified Partial

Failed

Unauthorized

Deviation Identified

Exception Active

Reassessment Required

Closed

Unknown

Conflicted

Each classification must remain connected to the transition scope, evidence, authority, and relevant time.

Governance Requirements

A conforming LTCM implementation must satisfy the following requirements. Transition Identity and Context

- Establish a stable transition case identity
- Identify the governed software object
- Establish the source lifecycle state
- Define the intended destination state
- Identify transition purpose
- Establish transition scope
- Preserve version and configuration applicability
- Preserve environment applicability
- Identify applicable specialized Parent Standards

Preconditions and Readiness

- Identify mandatory transition preconditions
- Establish verification requirements for preconditions
- Evaluate source-state validity
- Identify destination-state requirements
- Evaluate transition feasibility
- Establish transition readiness
- Identify relevant dependencies
- Preserve material uncertainty

Authority and Execution

- Identify competent transition authority
- Establish authority scope

• Identify execution responsibility

• Establish execution permission

• Define material execution boundaries

• Preserve transition decisions

• Maintain execution traceability

• Identify unauthorized transitions

• Prevent technical capability from substituting for authority

• Preserve tool-path authority where applicable

Outcome Verification

• Establish transition verification criteria

• Preserve execution evidence

• Verify the actual resulting state

• Distinguish execution from verification

• Identify partial transitions

• Identify failed transitions

• Identify interruptions

• Identify deviations

• Preserve residual obligations

• Classify transition outcomes

Continuity and Reassessment

• Establish recovery requirements where applicable

• Govern rollback as a distinct transition

• Preserve reversibility limitations

• Identify exception requirements

• Preserve automated execution boundaries

• Preserve autonomous execution authority

• Establish human intervention where required

• Support transition reassessment

• Preserve transition closure evidence

• Provide resulting state evidence to LSCCM and other applicable SLCS modules.

Governance Logic

Software Identity → Transition Case Identity → Source State → Source-State

Verification → Transition Purpose → Intended Destination State → Transition Scope → Applicable Requirements → Mandatory Preconditions → Feasibility → Readiness → Competent Authority → Execution Permission → Execution Boundaries → Governed Execution → Actual Resulting State → Outcome Verification → Deviation Assessment → Residual Obligations → Transition Classification → Transition Closure → Software Lifecycle Transition Control Profile

Where a transition is authorized and verified:

Verified Source State → Preconditions Satisfied → Competent Authority →

Governed Execution → Verified Destination State

Where execution is technically possible but unauthorized:

Transition Capability → Authority Assessment → Authority Absent →

Transition Not Authorized

Where execution occurs without legitimate authority:

Unauthorized Execution → Actual State Assessment → Deviation Governance →

Required Corrective Action

Where execution succeeds but verification is incomplete:

Transition Executed → Verification Gap → Verification Pending

Where a transition partially succeeds:

Intended Scope → Partial Execution → Actual State Reconciliation → Partial

Transition Classification

Where a transition fails:

Governed Execution → Failure → Actual State Assessment → Recovery or

Reassessment

Where an exception is required:

Transition Requirement → Exception Need → LEMM Governance → Applicable

Conditions → Transition Reassessment

LTCM must not treat successful technical execution as sufficient evidence of legitimate lifecycle transition.

Governance Outputs

The principal output is:

Software Lifecycle Transition Control Profile

Supporting outputs may include:

Lifecycle Transition Case Identity

Source-State Verification Record

Destination-State Definition

Transition Scope Profile

Transition Requirement Register

Transition Precondition Assessment

Transition Feasibility Assessment

Transition Readiness Profile

Transition Authority Record

Transition Execution Permission Record

Transition Execution Boundary Profile

Transition Execution Trace

Transition Verification Record

Transition Deviation Record

Transition Failure Record

Transition Recovery Record

Transition Rollback Record

Autonomous Transition Authority Record

Transition Outcome Classification

Transition Closure Record

Minimum Implementation Framework

A minimum LTCM implementation must establish eight governance layers. Layer 1 — Transition Identity

Identify the software object, transition case, source state, intended

destination state, and applicable scope. Layer 2 — Requirements and Preconditions

Identify the specialized governance requirements, mandatory preconditions,

dependencies, and verification criteria applicable to the transition. Layer 3 — Readiness Assessment

Evaluate source-state validity, feasibility, resource availability,

operational conditions, and transition readiness. Layer 4 — Authority and Permission

Establish competent decision authority, execution responsibility, permitted

actions, and material execution boundaries. Layer 5 — Governed Execution

Preserve traceability between the authorized transition and actual

execution activities. Layer 6 — Outcome Verification

Verify the resulting lifecycle state and distinguish complete, partial,

failed, unauthorized, and unverified outcomes. Layer 7 — Deviation and Recovery

Identify material deviations, interruptions, exception requirements,

rollback conditions, and recovery obligations. Layer 8 — Closure and Continuity

Preserve transition decisions, resulting state evidence, unresolved

obligations, reassessment triggers, and closure records.

The framework may be implemented through lifecycle management platforms, release systems, deployment tools, configuration management systems, workflow controls, or AI-assisted analysis.

The implementation mechanism must adapt to the applicable lifecycle governance requirements.

Cross-Module Boundary

LSCCM → LTCM

LSCCM establishes evidence-supported lifecycle state consistency.

LTCM governs legitimate movement between lifecycle states.

LTCM → LEMM

LTCM identifies transition requirements and exceptional conditions.

LEMM governs the legitimacy, authorization, restrictions, and continuity of

lifecycle exceptions.

LTCM → LCTCM

LTCM establishes transition authority and execution responsibility.

LCTCM governs the transfer and continuity of lifecycle control between

actors or systems.

LTCM → LCAM

LTCM provides evidence of transition control design and execution.

LCAM assesses whether the relevant lifecycle control mechanisms are suitable and effective.

Transition legitimacy is distinct from state consistency, exception authorization, control transfer, and overall control effectiveness.

Cross-Architecture Boundary

LTCM governs lifecycle transition control within SLGA™.

It does not replace the specialized mechanisms that establish requirements for individual transitions.

SRS — Software Requirements Standard

Requirements-related lifecycle transitions.

SCS — Software Creation Standard

Creation and development transitions.

SDS — Software Dependency Standard

Dependency introduction, replacement, and continuity conditions.

SCFS — Software Configuration Standard

Configuration, build, and artifact transitions.

SQS — Software Quality Standard

Quality assessment conditions relevant to lifecycle progression.

SSS — Software Security Standard

Security conditions, restrictions, and remediation requirements.

SRLS — Software Release Standard

Release readiness, authorization, and withdrawal.

SDPS — Software Deployment Standard

Deployment execution, operational state, and recovery.

SCMS — Software Change Management Standard

Change governance and lifecycle re-entry.

SRTS — Software Retirement Standard

Retirement planning, migration, withdrawal, dependency reconciliation, and

closure.

LTCM may also interact with applicable OOF® authority, accountability, identity, operational reality, AI, and autonomous systems governance mechanisms.

It does not redefine their specialized governance functions.

Practical Applications

Software Development Governance

Controlling transitions between development, validation, and

release-related states.

Enterprise Software

Establishing authorized lifecycle progression across teams, platforms, and

operational environments.

Cloud Deployment

Governing deployment, activation, restriction, recovery, and withdrawal

transitions.

Software Supply Chains

Controlling lifecycle transitions associated with dependency changes and

component replacement.

AI Systems

Governing transitions between development, evaluation, deployment,

operational use, restriction, and retirement.

Autonomous Software Agents

Preserving authority boundaries for automated or autonomous lifecycle

actions.

Industrial Software

Establishing legitimate transition conditions for software operating within physical equipment and production systems.

Critical Infrastructure

Preserving authorization, verification, and continuity during

high-consequence software transitions.

Practical Use Case 1 — Unauthorized Production Deployment

An organization uses an automated deployment pipeline to release software

into production.

A new software build passes its technical tests and becomes available for

deployment.

The pipeline is configured to permit deployment when technical checks succeed.

However, the applicable governance requirements also require a separate

production authorization decision.

An automated process initiates deployment without that authorization.

LTCM evaluates:

Source State → Destination State → Technical Readiness → Mandatory

Preconditions → Competent Authority → Execution Permission → Actual Deployment → Verification

The assessment establishes:

• The build satisfied the technical test requirements

• Production authorization was mandatory

• The required authorization was absent

• The deployment was technically executed

• The resulting software state must still be verified

• The transition contains an authority violation regardless of technical

success.

The transition is classified as:

Unauthorized Transition — Actual Destination State Assessment Required.

The relevant release, deployment, and change governance mechanisms

determine the corrective response.

Technical Readiness ≠ Production Authorization.

Practical Use Case 2 — Autonomous Agent Changes Software Lifecycle State

An organization deploys an autonomous software maintenance agent.

The agent is authorized to identify defects, prepare code changes, and initiate testing.

During operation, the agent determines that a modified software component should be deployed immediately.

It selects a deployment tool and attempts to move the component from a

testing environment into production.

LTCM evaluates:

Agent Objective → Assigned Authority → Proposed Transition → Source State

→ Destination State → Tool Selection → Execution Permission → Operational Boundary

The assessment identifies:

• The agent is authorized to prepare modifications

• The agent is authorized to initiate testing

• Production deployment requires separate authority

• The agent possesses technical access to a deployment tool

• Technical access does not extend its legitimate transition authority

The production transition must remain blocked or subject to the applicable authorization mechanism.

The agent may continue its authorized development and testing activities.

Authority to Modify Software ≠ Authority to Deploy Software.

Critical Distinctions

Source State Recorded ≠ Source State Verified.

Transition Proposed ≠ Transition Authorized.

Transition Required ≠ Transition Permitted.

Technical Feasibility ≠ Transition Readiness.

Transition Readiness ≠ Transition Authority.

Authority Established ≠ Execution Permission Unrestricted.

Technical Access ≠ Legitimate Execution Permission.

Task Authority ≠ Constituent Action Authority.

Transition Executed ≠ Transition Verified.

Destination State Reached ≠ Transition Legitimately Executed.

Transition Successful Technically ≠ Transition Compliant.

Partial Transition ≠ Complete Transition.

Transition Interrupted ≠ Transition Failed Permanently.

Rollback Available ≠ Transition Fully Reversible.

Rollback Executed ≠ Previous State Completely Restored.

Lifecycle Re-entry ≠ Complete Lifecycle Repetition.

Authorized Exception ≠ Elimination of Mandatory Governance.

Automated Execution ≠ Autonomous Authority.

Transition Case Closed ≠ Entire Lifecycle Activity Completed.

Canonical Boundary

Lifecycle Transition Control Module (LTCM) governs the legitimacy,

conditions, authority, execution boundaries, verification, and resulting disposition of material software lifecycle transitions.

It establishes the governance relationship between a verified source state,

an intended destination state, applicable requirements, competent authority, actual execution, and verified outcome.

It supports nonlinear, partial, automated, autonomous, conditional, and

exceptional lifecycle transitions without imposing one universal lifecycle sequence.

It does not replace specialized software creation, quality, security, release, deployment, change, or retirement governance.

It does not create authority through technical capability or methodological classification.

Its principal governance output is the:

Software Lifecycle Transition Control Profile

This output establishes what transition was intended, under what conditions

it was permitted, who or what executed it, what actually changed, what was verified, and which obligations remain.

Canonical Closing Statement

Software lifecycle transitions are not legitimate merely because software can move from one state to another.

A build may be technically ready without being authorized for release.

A deployment pipeline may be capable of changing production without possessing the required permission.

A software component may be modified without satisfying the conditions for operational use.

An autonomous agent may possess tools that allow actions beyond its assigned authority.

And a technically successful transition may still leave unresolved dependencies, incomplete verification, or continuing governance obligations.

These conditions demonstrate that software lifecycle control requires more than tracking movement between states.

LTCM establishes the governance mechanism that connects transition purpose, source conditions, destination requirements, legitimate authority, execution boundaries, and actual outcomes.

It prevents technical capability from being confused with permission.

It prevents execution records from substituting for verification.

It preserves the distinction between partial, failed, unauthorized, and

legitimately completed transitions.

It supports automation and autonomy without abandoning authority boundaries.

And it ensures that lifecycle transitions remain traceable to the conditions under which they were permitted and the operational states they actually produced.

A software lifecycle transition is not governed because it happened. It is governed when the right transition occurred, under legitimate authority, within defined boundaries, and with a verifiable result.

Lifecycle Transition Control Module (LTCM) establishes the authority and

verification foundation of software lifecycle movement—ensuring that software changes its governed state through legitimate transitions rather than through technical capability alone.